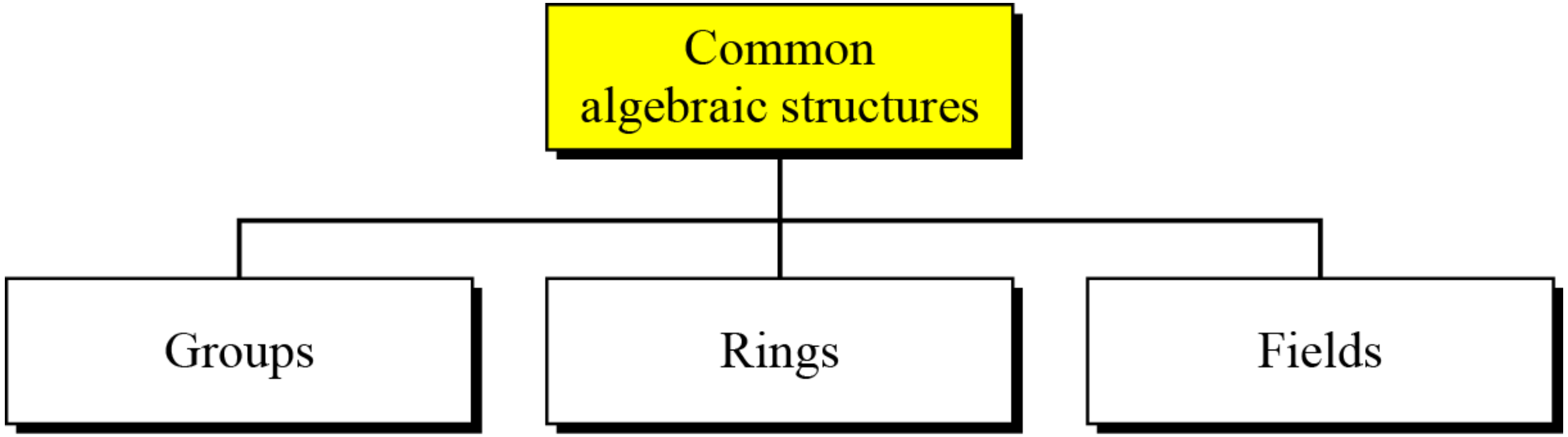
Algebraic Structure

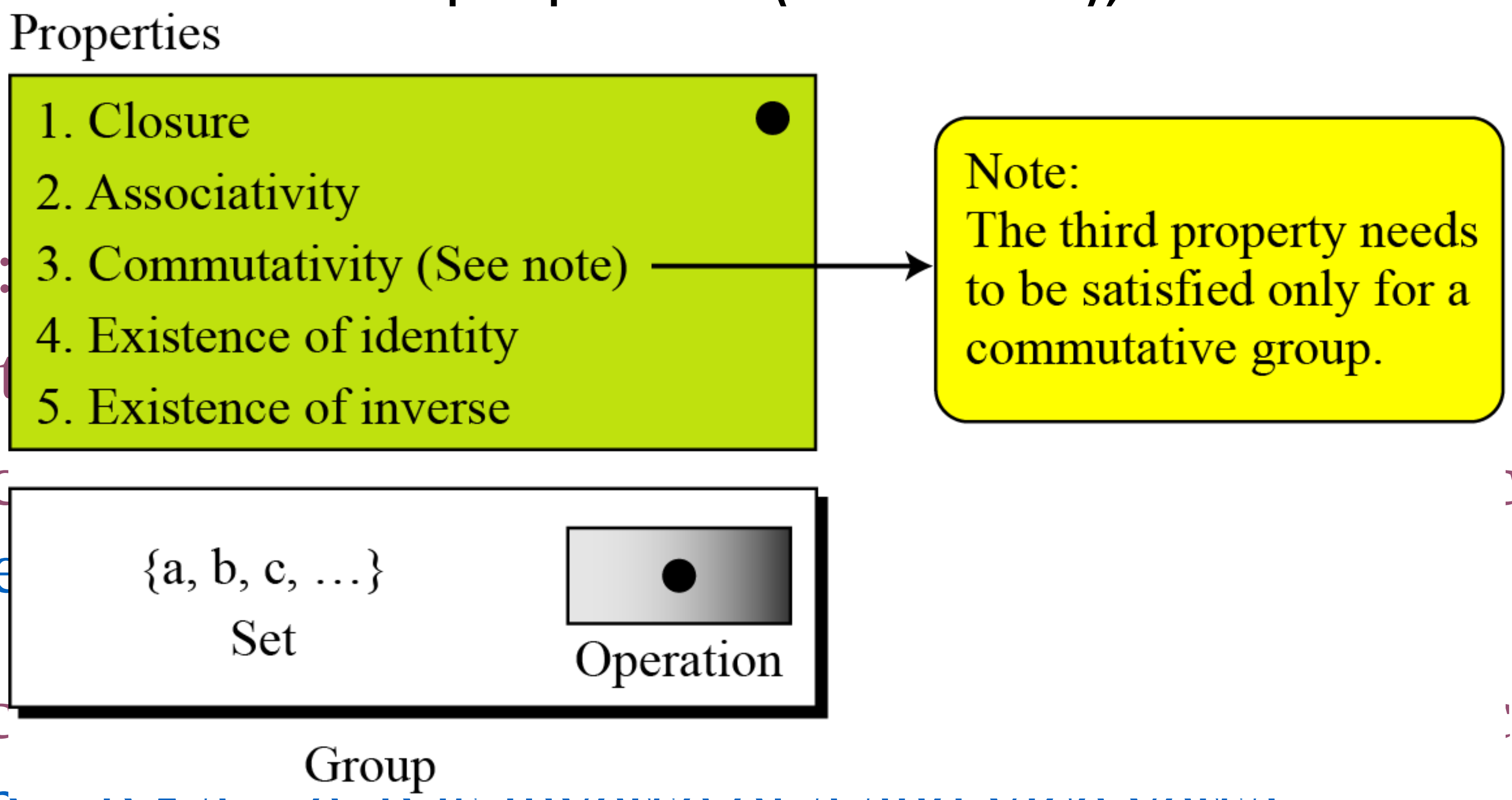
Algebraic Structures



- *Cryptography requires sets of integers and specific operations that are defined for those sets*
- *The **combination of the set and the operations** that are applied to the elements of the set is called an algebraic structure*
- *In this chapter, we will define three common algebraic structures: groups, rings, and fields.*

Groups

- A group (G) is a set of elements with a binary operation (•) that satisfies four properties (or axioms), denoted as (G, •)



Closure:

Associat

Existenc ch

that a • e

Existenc ch

that a • b = b • a = e, b is inverse of a and vice versa

…Groups

- A group (G, •) is called commutative or abelian group if the operator '•' satisfies the commutative property
  - Commutative property: for all a, b $\in$ G, a • b = b • a

Although a group involves a single operation, the properties imposed on the operation allow the use of a pair of operations as long as they are inverses of each other.

Example: groups

The set of residue integers with the addition operator,

$$G = < Z_n , +>,$$

is a commutative group. Also, We can perform addition and subtraction on the elements of this set without moving out of the set.

The set Z_n^* with the multiplication operator, $G = <Z_n^*, \times>$, is also an abelian (commutative) group.

Let us define a set $G = < \{a, b, c, d\}, \bullet>$ and the operation as shown in following table

every element x . y matches its mirror y . x across the diagonal, the group is confirmed to be abelian.

e.g. a.c=c

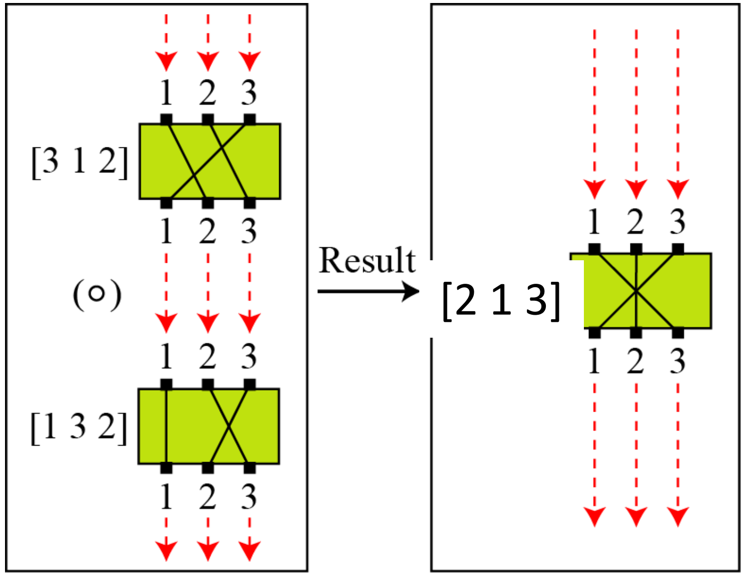
c.a=c

•	*a*	*b*	*c*	*d*
a	*a*	*b*	*c*	*d*
b	*b*	*c*	*d*	*a*
c	*c*	*d*	*a*	*b*
d	*d*	*a*	*b*	*c*

This is an abelian group (follows all properties)

a is the identity element.

Permutation group



Each green rectangle is a permutation machine

This is not an abelian group

∘	[1 2 3]	[1 3 2]	[2 1 3]	[2 3 1]	[3 1 2]	[3 2 1]
[1 2 3]	[1 2 3]	[1 3 2]	[2 1 3]	[2 3 1]	[3 1 2]	[3 2 1]
[1 3 2]	[1 3 2]	[1 2 3]	[2 3 1]	[2 1 3]	[3 2 1]	[3 1 2]
[2 1 3]	[2 1 3]	[3 1 2]	[1 2 3]	[3 2 1]	[1 3 2]	[2 3 1]
[2 3 1]	[2 3 1]	[3 2 1]	[1 3 2]	[3 1 2]	[1 2 3]	[2 1 3]
[3 1 2]	[3 1 2]	[2 1 3]	[3 2 1]	[1 2 3]	[2 3 1]	[1 3 2]
[3 2 1]	[3 2 1]	[2 3 1]	[3 1 2]	[1 3 2]	[2 1 3]	[1 2 3]

This is the permutation group on three symbols, The operation ∘ is composition of permutations.

…Permutation group

- set of permutations with the composition operation is a group
  - This implies that using two permutations one after another **cannot strengthen** the security of a cipher
    - because we can always find a permutation that can do the same job because of the **closure property**

Using permutations alone, no matter how many rounds, is equivalent to a single permutation.

This is why: **A pure transposition cipher or any cipher that only permutes symbols is cryptographically weak**.

…Groups

- **Finite Group**: a group G with finite elements (otherwise, infinite group)

Z_6={0,1,2,3,4,5} under addition mod 6 (Finite group -> 6 elements)

- **Order of a Group** |G|: Number of elements in the group (if finite); otherwise, infinity (∞)

$|Z_6| = 6$ and $|Z| = \infty$

- **Subgroup**: A subset H of G is a subgroup of G, if H itself is a group under the operation of G

Subgroups

- If $a, b \in H$ ➔ $c = a \bullet b \in H$
- $e \in G, H$
- If $a \in H$ ➔ $a^{-1} \in H$
- $(\{e\}, \bullet)$ is subgroup of G, H (The **set containing only the identity element is always a subgroup**)
- **G is a subgroup of itself**

Example: subgroup

Is the group $H = <Z_{10}, +>$ a subgroup of the group $G = <Z_{12}, +>$?

The answer is **no**. The operations defined for these two groups are different. The operation in H is addition modulo 10; the operation in G is addition modulo 12. Because the rules for adding elements differ (e.g., 7+5 is 2 in H, but 0 in G), H cannot be a subgroup of G.

Cyclic Subgroups

- If a subgroup of a group can be generated using the **power of an element**, the subgroup is called the cyclic subgroup

$$\boxed{a^n \rightarrow a \bullet a \bullet \ldots \bullet a \quad (n \text{ times})}$$

Example: cyclic subgroup

Four cyclic subgroups can be made from the group $G = \langle Z_6, +\rangle$. They are $H_1 = \{0\}$, $H_2 = \{0, 2, 4\}$, $H_3 = \{0, 3\}$, and $H_4 = Z_6$.

$0^0 \bmod 6 = 0$

$1^0 \bmod 6 = 0$
$1^1 \bmod 6 = 1$
$1^2 \bmod 6 = (1 + 1) \bmod 6 = 2$
$1^3 \bmod 6 = (1 + 1 + 1) \bmod 6 = 3$
$1^4 \bmod 6 = (1 + 1 + 1 + 1) \bmod 6 = 4$
$1^5 \bmod 6 = (1 + 1 + 1 + 1 + 1) \bmod 6 = 5$

$2^0 \bmod 6 = 0$
$2^1 \bmod 6 = 2$
$2^2 \bmod 6 = (2 + 2) \bmod 6 = 4$

$3^0 \bmod 6 = 0$
$3^1 \bmod 6 = 3$

$4^0 \bmod 6 = 0$
$4^1 \bmod 6 = 4$
$4^2 \bmod 6 = (4 + 4) \bmod 6 = 2$

$5^0 \bmod 6 = 0$
$5^1 \bmod 6 = 5$
$5^2 \bmod 6 = 4$
$5^3 \bmod 6 = 3$
$5^4 \bmod 6 = 2$
$5^5 \bmod 6 = 1$

…Example: cyclic subgroup

Three cyclic subgroups can be made from the group $G = \langle Z_{10}^*, \times\rangle$. G has only four elements: 1, 3, 7, and 9. The cyclic subgroups are $H_1 = \{1\}$, $H_2 = \{1, 9\}$, and $H_3 = Z_{10}^*$.

$1^0 \bmod 10 = 1$

$3^0 \bmod 10 = 1$
$3^1 \bmod 10 = 3$
$3^2 \bmod 10 = 9$
$3^3 \bmod 10 = 7$

$7^0 \bmod 10 = 1$
$7^1 \bmod 10 = 7$
$7^2 \bmod 10 = 9$
$7^3 \bmod 10 = 3$

$9^0 \bmod 10 = 1$
$9^1 \bmod 10 = 9$

Cyclic Groups

- A group G is cyclic **if there exists an element g** (called a **generator**) such that every element in G can be written as g^k (or k . g for additive groups) for some integer k.

$$\boxed{\{e, g, g^2, \ldots, g^{n-1}\}, \text{ where } g^n = e}$$

Cyclic Groups

As seen in the earlier examples:

a. The group G = $\langle Z_6, + \rangle$ is a cyclic group with two generators, $g = 1$ and $g = 5$.

b. The group G = $\langle Z_{10}^*, \times \rangle$ is a cyclic group with two generators, $g = 3$ and $g = 7$.

Subgroup property

Assume that G is a group, and H is a subgroup of G. If the order of G and H are |G| and |H|, respectively, then |H| divides |G|.

Let:

$$G = \mathbb{Z}_{12} = \{0, 1, 2, \ldots, 11\}$$

Take the subgroup:

$$H = \{0, 4, 8\}$$

- $|G| = 12$
- $|H| = 3$

$$12 = 4 \times 3$$

Order of an Element

The order of an element a, ord(a), is the smallest number n such that a^n =e (for multiplicative notation) or a*n =e (for additive notation)

In other words, ord(a) is the order of the cyclic group generated by a

Example: ord(a)

a. In the group $G = \langle Z_{10}^*, \times \rangle$, the orders of the elements are:

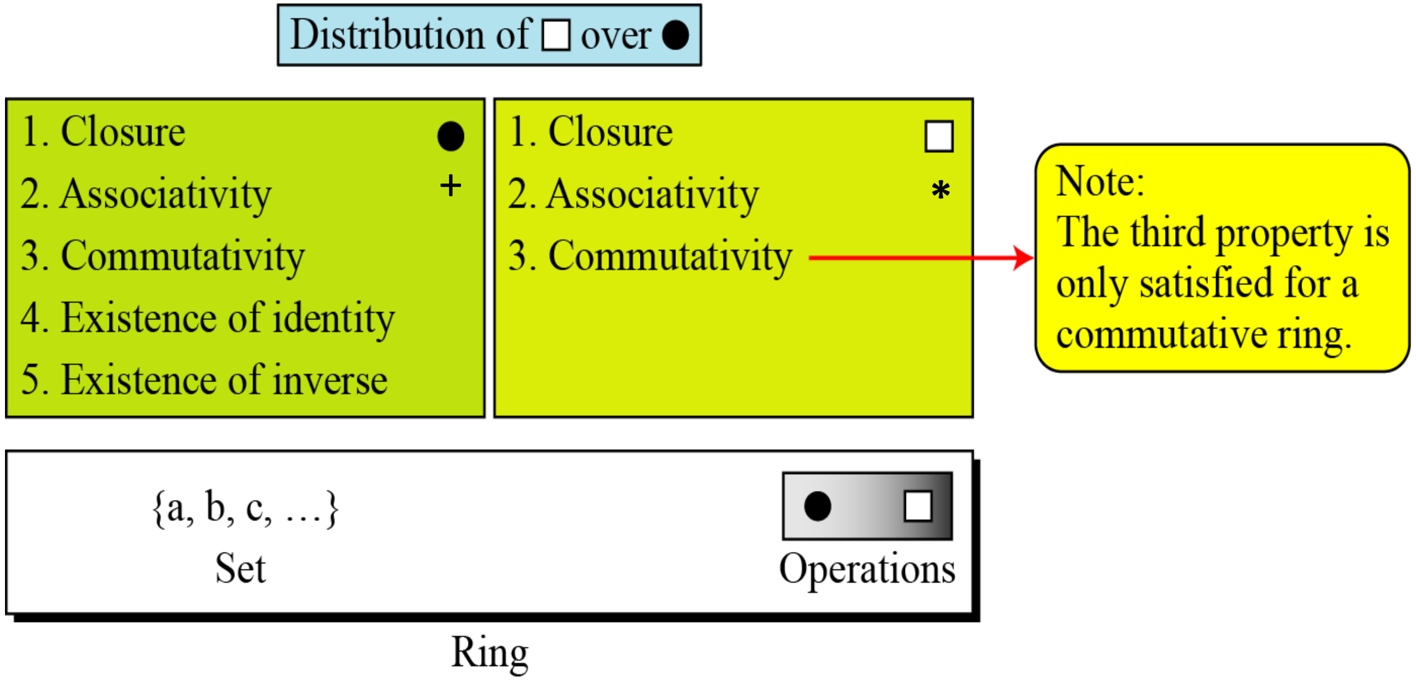
ord(1) = 1, ord(3) = 4, ord(7) = 4, ord(9) = 2. (e.g. for ord(3) => $3^4=81 \equiv 1 \bmod 10$ ord(7) => $7^4=2401 \equiv \mathbf{1} \bmod 10$)

b. In the group $G = \langle Z_6, + \rangle$, the orders of the elements are:

ord(0) = 1, ord(1) = 6, ord(2) = 3, ord(3) = 2, ord(4) = 3, ord(5) = 6. (e.g. for ord(2) => $2 \cdot \mathbf{3} = 6 \equiv \mathbf{0} \bmod 6$)

Ring

- A ring, R = <{…}, +,*>, is an **algebraic structure with two operations**
  - Respect to first operation '+' (also called addition) R is an abelian group
  - Respect to second operation (normally called multiplication) '*', R is semi group i.e., closure and associative rules are satisfied
  - '*' is distributive over '+'
    - a*(b+c) = a*b +a*c

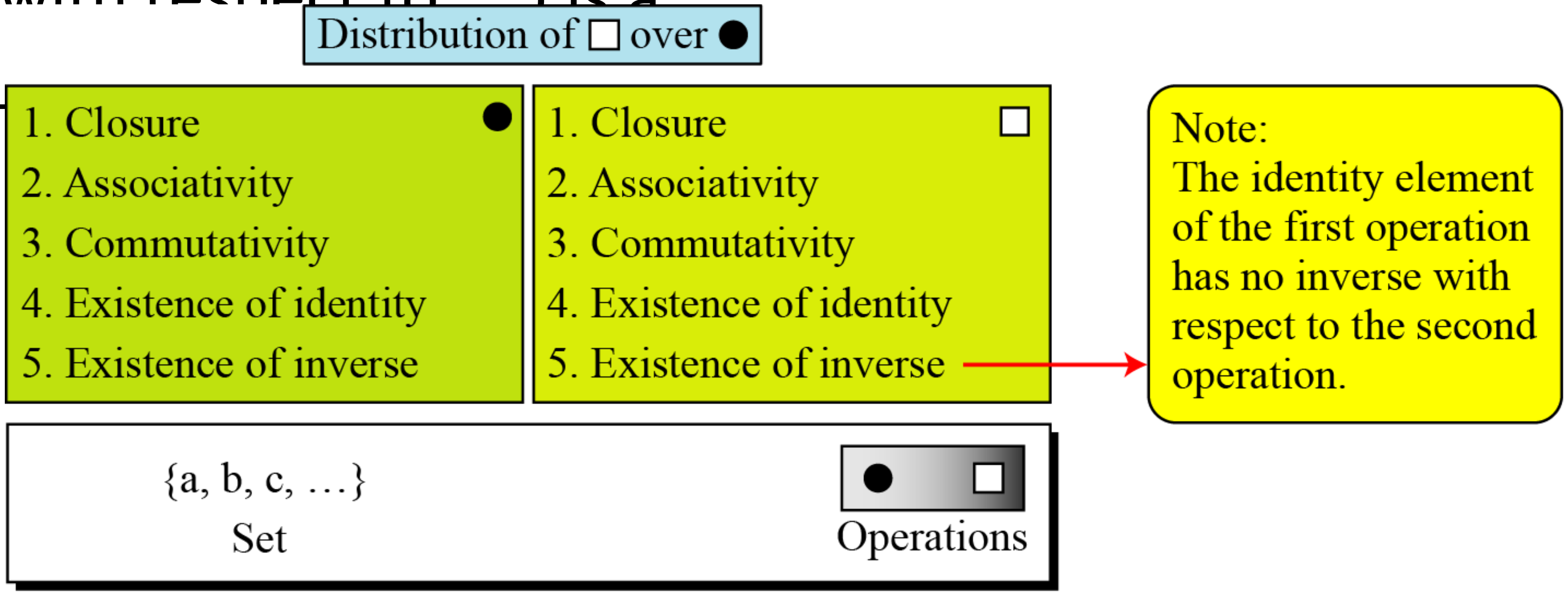


The set Z with two operations, addition and multiplication, is a **commutative ring**. We denote it by R = <Z, +, *>. Addition satisfies all of the five properties; multiplication satisfies only three properties.

If '*' is commutative, R is commutative ring

Field

- A field, denoted by F = <{…}, +, *>
- **{…}** is a <u>commutative group respect to the first operation ‘+’</u>
- **{…} - {0}** (as 0 is the identity of first operation ‘+’ has no inverse with respect to ‘*’) is a <u>commutative g</u>



Field

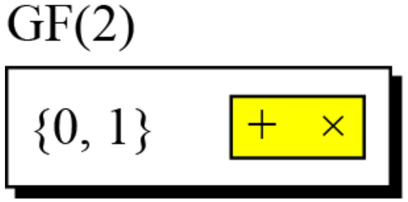
Finite field

- Finite field is important in cryptography
- A field with finite number is called finite field
- Galois showed that for a field to be finite, the number of elements should be p^n, where ***p* is a prime and *n* is a positive integer**

Galois field GF(p)

- When n = 1, we have GF(p) field. This field can be the set Z_p= {0, 1, …, p − 1}, with two arithmetic operations

A very common field in this category is GF(2) with the set {0, 1} and two operations, addition and multiplication, as shown in Figure.



+	0	1
0	0	1
1	1	0

Addition

×	0	1
0	0	0
1	0	1

Multiplication

Inverses

Addition: $0^{-1} = 0$ $1^{-1}=1$

Multiplication: 0^{-1} =NA $1^{-1}=1$

...Galois field GF(p)

We can define GF(5) on the set Z_5 (5 is a prime) with addition and multiplication operators as shown below

GF(5)

$\{0, 1, 2, 3, 4\}$ $+\ \times$

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplication

Additive inverse

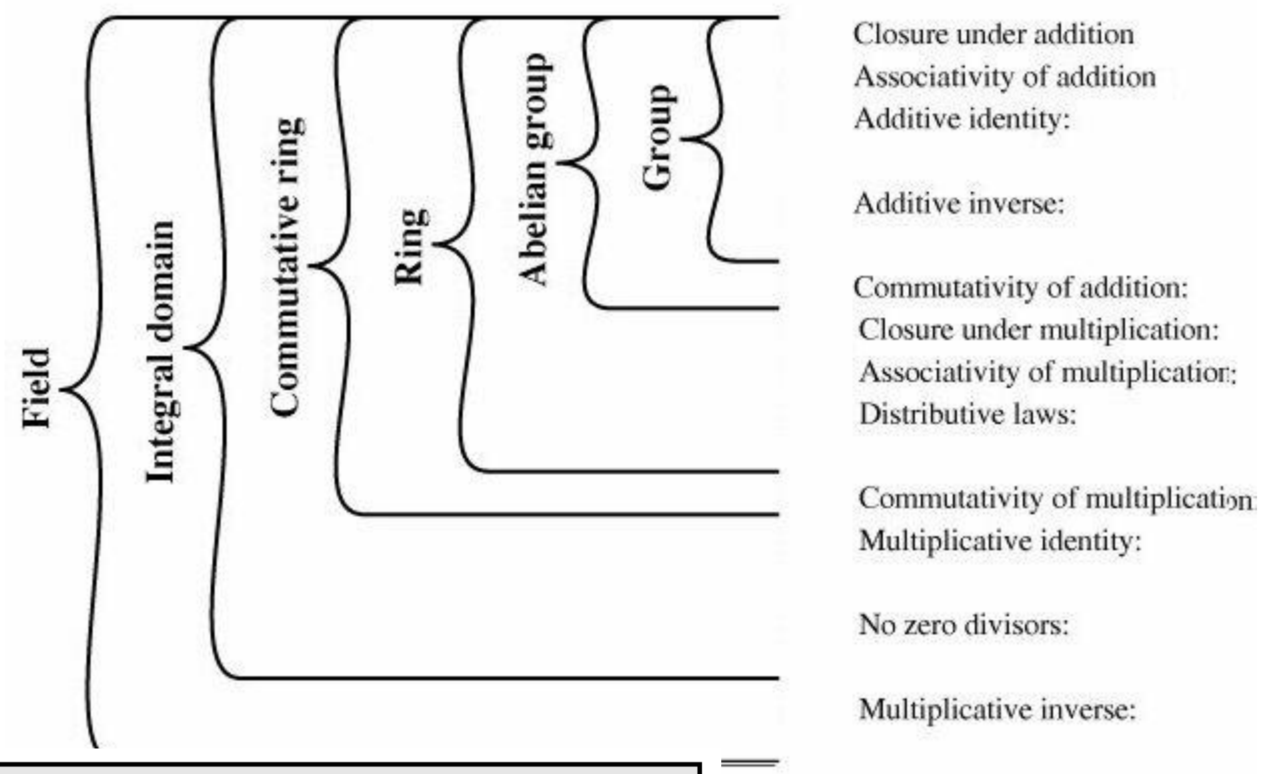
a	0	1	2	3	4
$-a$	0	4	3	2	1

a	0	1	2	3	4
a^{-1}	—	1	3	2	4

Multiplicative inverse

Cayley tables

Summary



Algebraic Structure	*Supported Typical Operations*	*Supported Typical Sets of Integers*
Group	(+ −) or (× ÷)	$\mathbf{Z}_n$ or $\mathbf{Z}_n{}^*$
Ring	(+ −) and (×)	$\mathbf{Z}$
Field	(+ −) and (× ÷)	$\mathbf{Z}_p$

GF(2^n)

- In cryptography, we **often need to use four operations** (addition, subtraction, multiplication, and division) i.e. we **need to use fields**
- Although **GF(p)** can be implemented using n-bit words, **only values from 0 to p−1 are valid field elements**
  - numbers between p and 2^n - 1 must be reduced modulo p and therefore cannot be directly represented or used in the field.
- On the other side, **in GF(2^n), we have a set of 2^n elements**
  - and every possible n-bit word in computer memory corresponds to a valid field element. No representation is wasted
  - That is why **GF(2^n) is preferred** in many cryptographic systems

…GF(2^n)

Let us define a GF(2^2) field in which the set has four 2-bit words: {00, 01, 10, 11}. We can redefine addition and multiplication for this field in such a way that all properties of these operations are satisfied

Addition

⊕	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

Identity: 00

Multiplication

⊗	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

Identity: 01

00, 01, 10, 11 cannot be considered as integer from 0 to 3

Rules for multiplication

1. Multiply polynomials normally
2. Reduce modulo m(x) = $x^2 + x + 1$
3. Use arithmetic mod 2 (so 1+1=0)

Addition and multiplication are defined in terms of polynomial

Polynomials

- A polynomial of degree $n - 1$ is an expression like

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x^1 + a_0x^0$$

where a_i is called coefficient of the i^{th} term.

- 8-bit word 10011001 represents as

n-bit word	1 0 0 1 1 0 0 1
Polynomial	$1x^7 + 0x^6 + 0x^5 + 1x^4 + 1x^3 + 0x^2 + 0x^1 + 1x^0$
First simplification	$1x^7 + 1x^4 + 1x^3 + 1x^0$
Second simplification	$x^7 + x^4 + x^3 + 1$

…Polynomials

- To find the 8-bit word related to the polynomial $x^5 + x^2 + x$
  - we first supply the omitted terms
  - Since n = 8, it means the polynomial is of degree 7
  - The expanded polynomial is

$$0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$$

  - Related 8-bit word is 00100110

Operations on polynomials

- Any operation on polynomial involves two operations:
  - operation on coefficients and
  - operations on two polynomials
- Operations on coefficients (0/1) use GF(2)
- Operations on polynomials need GF(2^n)

Modulus respect to polynomial

- **Addition of two polynomials** never create a polynomial out side the set
- **Multiplication** of two polynomials may generates a po

Degree	*Irreducible Polynomials*
1	$(x + 1), (x)$
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1), (x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1), (x^4 + x^3 + 1), (x^4 + x + 1)$
5	$(x^5 + x^2 + 1), (x^5 + x^3 + x^2 + x + 1), (x^5 + x^4 + x^3 + x + 1),$ $(x^5 + x^4 + x^3 + x^2 + 1), (x^5 + x^4 + x^2 + x + 1)$

Addition operation

- Addition (or subtraction) over GF(2)
- $(x^5 + x^2 + x) \oplus (x^3 + x^2 + 1)$ in $GF(2^8)$, the symbol $\oplus$ to show that we mean polynomial addition

$$
\begin{array}{lcl}
0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0 & \oplus & \\
0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x^1 + 1x^0 & & \\
\text{---} & & \\
0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x^1 + 1x^0 & \rightarrow & x^5 + x^3 + x + 1
\end{array}
$$

Additive **identity**: **zero polynomial** because f(x)+0=f(x)

Additive **inverse**: **polynomial itself** because f(x)+f(x)=0

Multiplication Operation

1. The coefficient multiplication is done in GF(2).

2. The multiplying x^i by x^j results in x^{i+j}.

3. The multiplication **may create terms with degree more than $n - 1$**, which means the result needs to be reduced using a modulus polynomial.

Multiplication: example

Find the result of $(x^5 + x^2 + x) \otimes (x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$, the symbol $\otimes$ to represent multiplication of two polynomials

Solution

$$P_1 \otimes P_2 = x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$

$P_1 \otimes P_2 = x^{12} + x^9 + x$ … x^2

$P_1 \otimes P_2 = (x^{12} + x^7 + x$ …

Divisor: $x^8 + x^4 + x^3 + x + 1$

Quotient: $x^4 + 1$

$$
\begin{array}{l}
x^{12} + x^7 + x^2 \\
x^{12} + x^8 + x^7 + x^5 + x^4 \\
\hline
x^8 + x^5 + x^4 + x^2 \\
x^8 + x^4 + x^3 + x + 1 \\
\hline
\end{array}
$$

Remainder $\boxed{x^5 + x^3 + x^2 + x + 1}$

To find the final … by the polynomial of deg… …nder

Algorithm for multiplication

Another way to multiply $P_1 = (x^5 + x^2 + x)$ with $P_2 = (x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$

P_1 X P_2 = $x^5P_2 + x^2P_2 + xP_2$ mod IRP

Multiply P_2 by x, x^2, x^3,…

Powers	*Operation*	*New Result*	*Reduction*
$x^0 \otimes P_2$		$x^7 + x^4 + x^3 + x^2 + x$	No
$x^1 \otimes P_2$	$\boldsymbol{x} \otimes (x^7 + x^4 + x^3 + x^2 + x)$	$x^5 + x^2 + x + 1$	**Yes**
$x^2 \otimes P_2$	$\boldsymbol{x} \otimes (x^5 + x^2 + x + 1)$	$x^6 + x^3 + x^2 + x$	No
$x^3 \otimes P_2$	$\boldsymbol{x} \otimes (x^6 + x^3 + x^2 + x)$	$x^7 + x^4 + x^3 + x^2$	No
$x^4 \otimes P_2$	$\boldsymbol{x} \otimes (x^7 + x^4 + x^3 + x^2)$	$x^5 + x + 1$	**Yes**
$x^5 \otimes P_2$	$\boldsymbol{x} \otimes (x^5 + x + 1)$	$x^6 + x^2 + x$	No
$\mathbf{P_1} \times \mathbf{P_2} = (x^6 + x^2 + x) + (x^6 + x^3 + x^2 + x) + (x^5 + x^2 + x + 1) = \boldsymbol{x^5 + x^3 + x^2 + x + 1}$			

- Multiplication by *x* can be achieved by one bit left shift of P_2
- Need to be reduced after multiplication if degree greater than n-1
  - i.e., previously degree was n-1 (leading bit was 1) then reduce after multiplication result (XOR-ed with IRP)

Algorithm

- if leading bit of previous result '0'
  - One left shift
- if leading bit of previous result '1'
  - One left shift
  - XOR the result with least n-1 bits of IRP

P1 = 00100110,

P2 = 10011110,

IRP = 100011011

Remove 8th bit from IRP. 100011011 >> 00011011

Powers	*Shift-Left Operation*	*Exclusive-Or*
$x^0 \otimes P_2$		10011110
$x^1 \otimes P_2$	00111100	(00111100) $\oplus$ 00011011 = **<u>00100111</u>**
$x^2 \otimes P_2$	01001110	**<u>01001110</u>**
$x^3 \otimes P_2$	10011100	10011100
$x^4 \otimes P_2$	00111000	(00111000) $\oplus$ 00011011 = 00100011
$x^5 \otimes P_2$	01000110	**<u>01000110</u>**
$P_1 \otimes P_2$ = (00100111) $\oplus$ (01001110) $\oplus$ (01000110) = 00101111		

$$x^5 + x^3 + x^2 + x + 1$$

P_1 X P_2 = $x^5P_2 + x^2P_2 + xP_2$ mod IRP

Note: A field GF(2^n) may have more than one irreducible polynomials

Using generator

- It is **easier to define the elements of GF(2^n) using a generator**
- A generator is a special element of the field that acts like a "root" of the irreducible polynomial.
  - **If your irreducible polynomial is f(x), then g is defined such that f(g) = 0**
- The **complete set of elements in GF(2^n) using this generator is {0, g^0, g^1, g^2, …, g^N}**, where $N = 2^n - 2$. e.g. for GF(2^4) $N=2^4-2=14$ … let's see on the next slide

Elements of GF(2^4): IRP $g^4 + g + 1$

- $g^4 + g + 1 = 0$ ➔ $g^4 = g + 1$ (GF(2) rule: Addition = subtraction)
- The elements 0, g^0, g^1, g^2, and g^3 can be easily generated
- … d to

g^5 mod $g^4 + g + 1 = g^2$+g

0	=	0	=	0	=	0	$\longrightarrow$	0	=	(0000)
g^0	=	g^0	=	g^0	=	g^0	$\longrightarrow$	g^0	=	(0001)
g^1	=	g^1	=	g^1	=	g^1	$\longrightarrow$	g^1	=	(0010)
g^2	=	g^2	=	g^2	=	g^2	$\longrightarrow$	g^2	=	(0100)
g^3	=	g^3	=	g^3	=	g^3	$\longrightarrow$	g^3	=	(1000)
g^4	=	g^4	=	g^4	=	$g + 1$	$\longrightarrow$	g^4	=	(0011)
g^5	=	$g\,(g^4)$	=	$g\,(g + 1)$	=	$g^2 + g$	$\longrightarrow$	g^5	=	(0110)
g^6	=	$g\,(g^5)$	=	$g\,(g^2 + g)$	=	$g^3 + g^2$	$\longrightarrow$	g^6	=	(1100)
g^7	=	$g\,(g^6)$	=	$g\,(g^3 + g^2)$	=	$g^3 + g + 1$	$\longrightarrow$	g^7	=	(1011)
g^8	=	$g\,(g^7)$	=	$g\,(g^3 + g + 1)$	=	$g^2 + 1$	$\longrightarrow$	g^8	=	(0101)
g^9	=	$g\,(g^8)$	=	$g\,(g^2 + 1)$	=	$g^3 + g$	$\longrightarrow$	g^9	=	(1010)
g^{10}	=	$g\,(g^9)$	=	$g\,(g^3 + g)$	=	$g^2 + g + 1$	$\longrightarrow$	g^{10}	=	(0111)
g^{11}	=	$g\,(g^{10})$	=	$g\,(g^2 + g + 1)$	=	$g^3 + g^2 + g$	$\longrightarrow$	g^{11}	=	(1110)
g^{12}	=	$g\,(g^{11})$	=	$g\,(g^3 + g^2 + g)$	=	$g^3 + g^2 + g + 1$	$\longrightarrow$	g^{12}	=	(1111)
g^{13}	=	$g\,(g^{12})$	=	$g\,(g^3 + g^2 + g + 1)$	=	$g^3 + g^2 + 1$	$\longrightarrow$	g^{13}	=	(1101)
g^{14}	=	$g\,(g^{13})$	=	$g\,(g^3 + g^2 + 1)$	=	$g^3 + 1$	$\longrightarrow$	g^{14}	=	(1001)

Inverse

- Additive inverse: the element itself e.g. $(g^4 + g + 1) + (g^4 + g + 1) = 0$
- Multiplicative inverse: for g^i, it is g^{-i} where $-i \equiv k \bmod 2^n - 1$

  e.g. Inverse of g^4 is $g^{-4} = g^{15-4} = g^{11}$

  (Class assignment: verify $g^4 \cdot g^{11} = 1$)

Operations

a. $g^3 + g^{12} + g^7 = g^3 + (g^3 + g^2 + g + 1) + (g^3 + g + 1) = g^3 + g^2 \rightarrow (1100)$

b. $g^3 - g^6 = g^3 + g^6 = g^3 + (g^3 + g^2) = g^2 \rightarrow (0100)$

0	$=$	0	$=$	0	$=$	0	$\longrightarrow$	0	$=$	(0000)
g^0	$=$	g^0	$=$	g^0	$=$	g^0	$\longrightarrow$	g^0	$=$	(0001)
g^1	$=$	g^1	$=$	g^1	$=$	g^1	$\longrightarrow$	g^1	$=$	(0010)
g^2	$=$	g^2	$=$	g^2	$=$	g^2	$\longrightarrow$	g^2	$=$	(0100)
g^3	$=$	g^3	$=$	g^3	$=$	g^3	$\longrightarrow$	g^3	$=$	(1000)
g^4	$=$	g^4	$=$	g^4	$=$	$g + 1$	$\longrightarrow$	g^4	$=$	(0011)
g^5	$=$	$g\,(g^4)$	$=$	$g\,(g + 1)$	$=$	$g^2 + g$	$\longrightarrow$	g^5	$=$	(0110)
g^6	$=$	$g\,(g^5)$	$=$	$g\,(g^2 + g)$	$=$	$g^3 + g^2$	$\longrightarrow$	g^6	$=$	(1100)
g^7	$=$	$g\,(g^6)$	$=$	$g\,(g^3 + g)$	$=$	$g^3 + g + 1$	$\longrightarrow$	g^7	$=$	(1011)
g^8	$=$	$g\,(g^7)$	$=$	$g\,(g^3 + g + 1)$	$=$	$g^2 + 1$	$\longrightarrow$	g^8	$=$	(0101)
g^9	$=$	$g\,(g^8)$	$=$	$g\,(g^2 + 1)$	$=$	$g^3 + g$	$\longrightarrow$	g^9	$=$	(1010)
g^{10}	$=$	$g\,(g^9)$	$=$	$g\,(g^3 + g)$	$=$	$g^2 + g + 1$	$\longrightarrow$	g^{10}	$=$	(0111)
g^{11}	$=$	$g\,(g^{10})$	$=$	$g\,(g^2 + g + 1)$	$=$	$g^3 + g^2 + g$	$\longrightarrow$	g^{11}	$=$	(1110)
g^{12}	$=$	$g\,(g^{11})$	$=$	$g\,(g^3 + g^2 + g)$	$=$	$g^3 + g^2 + g + 1$	$\longrightarrow$	g^{12}	$=$	(1111)
g^{13}	$=$	$g\,(g^{12})$	$=$	$g\,(g^3 + g^2 + g + 1)$	$=$	$g^3 + g^2 + 1$	$\longrightarrow$	g^{13}	$=$	(1101)
g^{14}	$=$	$g\,(g^{13})$	$=$	$g\,(g^3 + g^2 + 1)$	$=$	$g^3 + 1$	$\longrightarrow$	g^{14}	$=$	(1001)

Operations (contd…)

a. $g^9 \times g^{11} = g^{20} = g^{20 \bmod 15} = g^5 = g^2 + g \rightarrow (0110)$

b. $g^3 / g^8 = g^3 \times g^7 = g^{10} = g^2 + g + 1 \rightarrow (0111)$

0	=	0	=	0	=	0	$\longrightarrow$	0	=	(0000)
g^0	=	g^0	=	g^0	=	g^0	$\longrightarrow$	g^0	=	(0001)
g^1	=	g^1	=	g^1	=	g^1	$\longrightarrow$	g^1	=	(0010)
g^2	=	g^2	=	g^2	=	g^2	$\longrightarrow$	g^2	=	(0100)
g^3	=	g^3	=	g^3	=	g^3	$\longrightarrow$	g^3	=	(1000)
g^4	=	g^4	=	g^4	=	$g + 1$	$\longrightarrow$	g^4	=	(0011)
g^5	=	$g\,(g^4)$	=	$g\,(g + 1)$	=	$g^2 + g$	$\longrightarrow$	g^5	=	(0110)
g^6	=	$g\,(g^5)$	=	$g\,(g^2 + g)$	=	$g^3 + g^2$	$\longrightarrow$	g^6	=	(1100)
g^7	=	$g\,(g^6)$	=	$g\,(g^3 + g)$	=	$g^3 + g + 1$	$\longrightarrow$	g^7	=	(1011)
g^8	=	$g\,(g^7)$	=	$g\,(g^3 + g + 1)$	=	$g^2 + 1$	$\longrightarrow$	g^8	=	(0101)
g^9	=	$g\,(g^8)$	=	$g\,(g^2 + 1)$	=	$g^3 + g$	$\longrightarrow$	g^9	=	(1010)
g^{10}	=	$g\,(g^9)$	=	$g\,(g^3 + g)$	=	$g^2 + g + 1$	$\longrightarrow$	g^{10}	=	(0111)
g^{11}	=	$g\,(g^{10})$	=	$g\,(g^2 + g + 1)$	=	$g^3 + g^2 + g$	$\longrightarrow$	g^{11}	=	(1110)
g^{12}	=	$g\,(g^{11})$	=	$g\,(g^3 + g^2 + g)$	=	$g^3 + g^2 + g + 1$	$\longrightarrow$	g^{12}	=	(1111)
g^{13}	=	$g\,(g^{12})$	=	$g\,(g^3 + g^2 + g + 1)$	=	$g^3 + g^2 + 1$	$\longrightarrow$	g^{13}	=	(1101)
g^{14}	=	$g\,(g^{13})$	=	$g\,(g^3 + g^2 + 1)$	=	$g^3 + 1$	$\longrightarrow$	g^{14}	=	(1001)

$(g^k)^{-1} = g^{15-k}$

Therefore, $g^3 / g^8 = g^3 \times (g^8)^{-1} = g^3 \times g^7 = g^{10}$

Multiplicative Inverse

- Multiplicative identity: 1 i.e., 00000…0001
- Multiplicative inverse: **extended Euclidean algorithm** on the given polynomial and modulus polynomial

  In GF (2^4), find the inverse of $(x^2 + 1)$ modulo $(x^4 + x + 1)$.

Finding inverse means: we are looking for a polynomial that, when multiplied by $(x^2 + 1)$ results in a remainder of 1. i.e. **$(x^2 + 1)$*Inverse $\equiv$ 1 mod $(x^4 + x + 1)$**

Extended Euclid's Algorithm for Polynomials

(x^2+1)*Inverse ≡ 1 mod (x^4+x+1)

- We start with our two main polynomials in r1 and r2.
- The t columns are "trackers." We initialize them as 0 and 1 because 0 .r1 + 1 .r2 =r2.

Step 1: $(x^4+x+1) = (x^2+1)(x^2+1)+x$

Step 2: $(x^2+1)=(x)(x)+1$

Step 3: $x=(1)(x)+0$

r1=q . r2+r

q	r_1	r_2	r	t_1	t_2	t
(x^2+1)	(x^4+x+1)	(x^2+1)	(x)	(0)	(1)	(x^2+1)
(x)	(x^2+1)	(x)	(1)	(1)	(x^2+1)	(x^3+x+1)
(x)	(x)	(1)	(0)	(x^2+1)	(x^3+x+1)	(0)
	(1)	(0)		(x^3+x+1)	(0)	

$t=t_1-q.\ t_2$For GF(2), its same as **$t=t_1+q.\ t_2$**

$(x^4+x+1) \div (x^2+1)$ remainder is x

$(x^2+1) \div x$ remainder is 1

x ÷ 1 **remainder is 0 (Terminate now)**

Result is t1= (x^3+x+1)

Note: In the last iteration *value of t* is mentioned as 0 because it's the 'stop' sign after getting remainder as 0

Class work: Verify whether $(x^2+1)(x^3+x+1) \equiv 1 \bmod (x^4+x+1)$

…Multiplicative Inverse

In GF(2^8), find the inverse of (x^5) modulo ($x^8 + x^4 + x^3 + x + 1$).

q	r_1	r_2	r	t_1	t_2	t
(x^3)	$(x^8 + x^4 + x^3 + x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	(0)	(1)	(x^3)
$(x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$
(x)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$
$(x^3 + x^2 + 1)$	$(x^3 + x^2 + 1)$	(1)	(0)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$	(0)
	(1)	(0)		$(x^5 + x^4 + x^3 + x)$	(0)	

Class Assignment

Use of Galois Field in AES, Why?

- Numbers stay small
  - **everything in GF(2^8) is always one byte**, even after multiplication or division.
- **Inverses exist for each operation, which makes decryption possible**.
- AES combines division in the first half of the S box, a transpose in the row shift, and **matrix operations during MixColumns and the second half of the S box.** [SubBytes, ShiftRows, MixColumns, AddroundKey]
- In AES the **irreducible polynomial used is $X^8 + X^4 + X^3 + X + 1$**

AES S-box

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

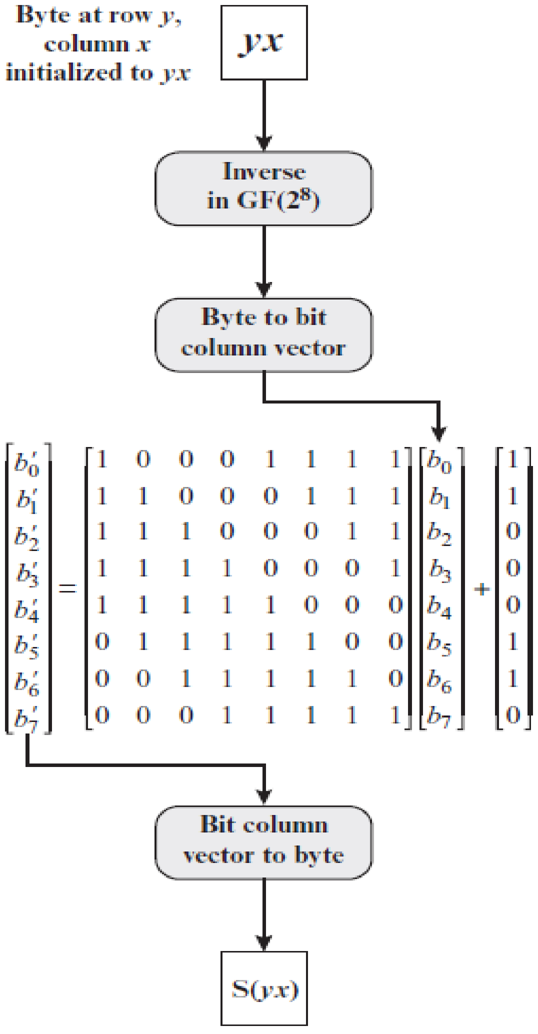
(a) S-box

AES Inverse S-Boxes

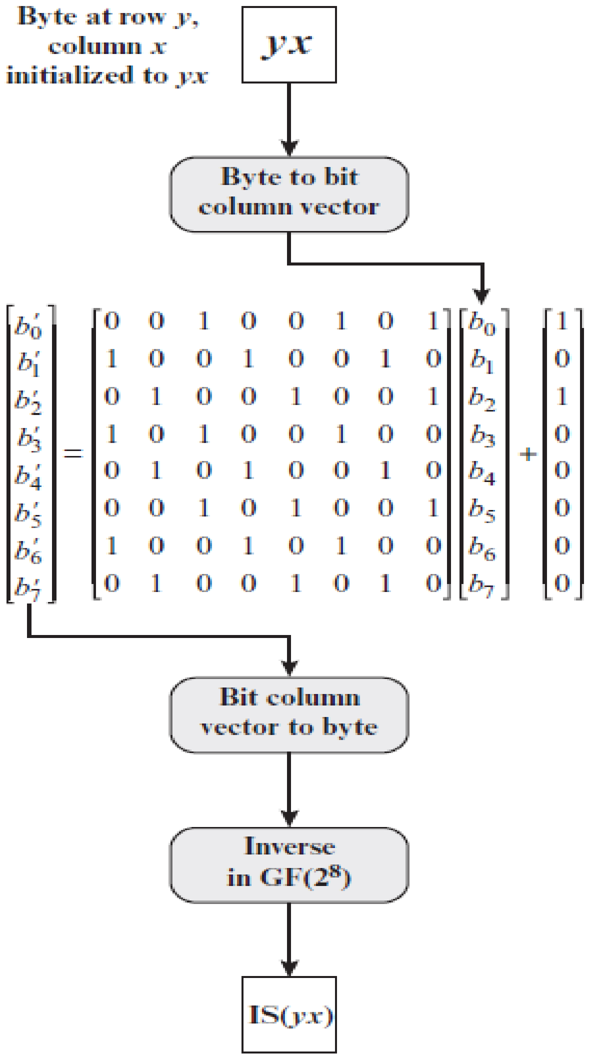
		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
	0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB
	1	7C	E3	39	82	9B	2F	FF	87	34	8E	43	44	C4	DE	E9	CB
	2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	0B	42	FA	C3	4E
	3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25
	4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92
	5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84
	6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06
x	7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	6B
	8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73
	9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF	6E
	A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B
	B	FC	56	3E	4B	C6	D2	79	20	9A	DB	C0	FE	78	CD	5A	F4
	C	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F
	D	60	51	7F	A9	19	B5	4A	0D	2D	E5	7A	9F	93	C9	9C	EF
	E	A0	E0	3B	4D	AE	2A	F5	B0	C8	EB	BB	3C	83	53	99	61
	F	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	0C	7D

(b) Inverse S-box

Construction of S-Box



(a) Calculation of byte at row y, column x of S-box



(a) Calculation of byte at row y, column x of IS-box

Recall of Cayley tables

GF(5)

{0, 1, 2, 3, 4} + ×

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition